Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Driving force

Meaning in my words How would you explain driving force to a friend?	**How can you tell?** A rope pulls right while friction acts left. Which one is the driving force on the cart?
Example or sketch Draw a 2 N driving pull and 0.4 N of friction. Which arrow is the driving force?	**Non-example and why** Is the hanging weight always the same as the rope tension on the cart?

Total moving mass

Meaning in my words How would you explain total moving mass to a friend?	**How can you tell?** If you move a washer from the cart onto its hanger, what happens to total moving mass?
Example or sketch A 0.6 kg cart side and 0.1 kg hanger side move together. What total should you use?	**Non-example and why** A student adds a new washer from the bench to the hanger. Is moving mass still controlled?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Proportional

Meaning in my words What does it mean to say acceleration is proportional to net force?	**How can you tell?** At fixed mass, if net force triples, what happens to acceleration?
Example or sketch For a 2 kg cart, compare the accelerations under 2 N and 4 N net.	**Non-example and why** Two readings both rise, but one doubles and the other barely changes. Is that enough to call them proportional?

Friction offset

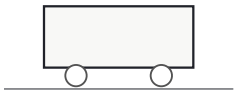
Meaning in my words How would you explain friction offset to a friend?	**How can you tell?** If the pull is 3 N and opposing friction is 1 N, how much force accelerates the cart?
Example or sketch For a sliding 2 kg cart with 1 N opposing friction, what acceleration would a 3 N pull produce?	**Non-example and why** A student uses the full 3 N pull as the net force. What did they leave out?

Doodle the idea

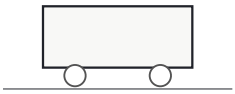
SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch cart and hanger before and after the transfer. Label the moving mass and the changed driving force.

Before transfer



After transfer



My marks mean

Which run changes more?

12

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when net forces differ.

CHOOSE ONE JOB TOGETHER

FLIGHT-TEST ENGINEER ☐

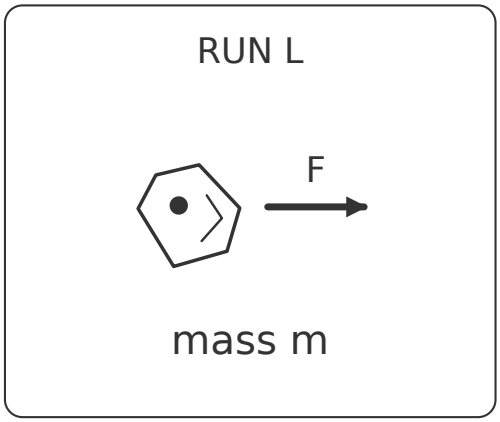
Add velocity-change arrows for both runs. Recommend the larger change and name what stayed the same.

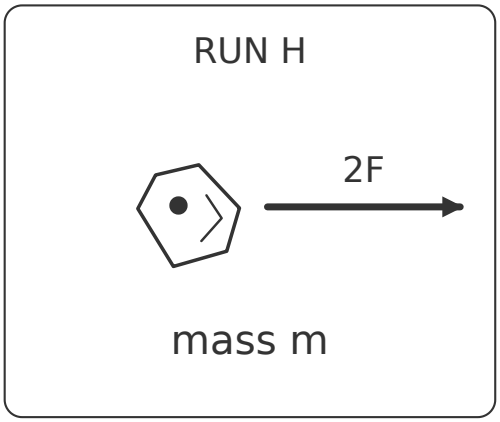
TEST-RANGE BROADCASTER ☐

Tell viewers which run changes velocity more. Ask whether doubling motor pull always doubles the net force.

Point to a clue. Keep your first marks; add the update using the color key.

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.





01 OUR FIRST CALL

How do the two velocity changes compare? Point to the force arrows.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Do the new readings still support your comparison? Explain what you must check.

Switch speakers. Do those arrows show net force or just motor pull?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks